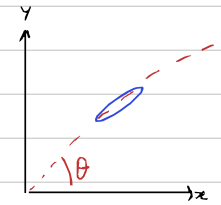
Predict Coast-Phase Apogee

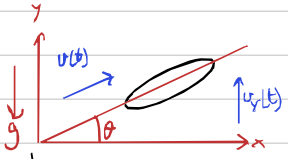
Apogee Height

$$\frac{dv}{dt} = -g - kv^2$$
$$\frac{dv}{dt} = v\frac{dv}{dh}$$
$$-g - kv^2 = v\frac{dv}{dh}$$
$$dh = v\,dv\,(-g - kv^2)^{-1}$$
$$dh = \frac{-v}{g+kv^2}\,dv$$



Integrating from initial velocity (v_0) to zero...

$$\int_{h_0}^{h_a} dh = \int_{v_0}^{0} \frac{-v}{g+kv^2}\,dv$$
$$h_a - h_0 = \int_{v_0}^{0} -\frac{v}{g+kv^2}\,dv$$
$$h_a = -\frac{1}{2k}\ln(g+kv^2)\Big|_{v_0}^{0} + h_0$$
$$h_a = -\frac{1}{2k}\ln(g+k(0)) - -\frac{1}{2k}\ln(g+kv_0^2) + h_0$$
$$h_a = \frac{1}{2k}\left[\ln\left(\frac{1}{g}\right) + \ln(g+kv_0^2)\right] + h_0$$
$$h_a = \frac{1}{2k}\left[\ln\left(\frac{g+kv_0^2}{g}\right)\right] + h_0$$
$$h_a = \frac{1}{2k}\ln\left(1+\frac{kv_0^2}{g}\right) + h_0$$



Find appropriate value for k.

$$k = \frac{1}{2m}\rho C_D S$$

Using RK4 to back-calculate from the known apogee value of 3450, mass was found to be approximately 36 kg. At 3450 m, $\rho \approx 0.868\ kg.m^{-3}$, $C_D = 0.4$ & $S = 0.0182\ m^2$

$$\Rightarrow k = \left(\frac{1}{2}\right)(36)(0.868)(0.4)(0.0182)$$
$$k \approx 8.8\times10^{-5}$$
$$\therefore H = \frac{1}{2k}\cdot\ln\left(1+\frac{k\times(280)^2}{9.81}\right) + \frac{1800}{3.281}$$
$$H \approx 3576\ m$$

Time to Apogee

$$dt = \frac{-dv}{g+kv^2}$$

Integrate from 0 to t_a...

$$\int_0^{t_a} dt = \int_{v_0}^{0} \frac{-dv}{g+kv^2}$$
$$t_a = \frac{-1}{\sqrt{gk}}\arctan\left(v\sqrt{\frac{k}{g}}\right)\Big|_{v_0}^{0}$$
$$t_a = \frac{1}{\sqrt{gk}}\arctan\left(v_0\sqrt{k/g}\right)$$

With the same values as before...

$$t_a = \frac{1}{\sqrt{9.81\times k}}\arctan\left(280\sqrt{\frac{k}{9.81}}\right)$$
$$t_a \approx 23.76\ \text{seconds after burnout}$$